MILITARY SPECIFICATION

# INSECTICIDE, D-PHENOTHRIN (AEROSOL)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Department and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of aerosol insecticide formulation containing d-phenothrin in a 12-ounce aerosol can complete with valve-actuator. An additional actuator is supplied with the aerosol can.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| BB-F-1421 | - | Fluorocarbon Refrigerants |
| NN-P-71 | - | Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial) |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |
| PPP-C-96 | - | Cans, Metal, 28 Gage and Lighter |

FSC 6840

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act
49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "40 CFR 162" or "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D3069 - Delivery Rate of Aerosol Products
D3091 - Particle-Size Distribution of Space-Insecticide Aerosol

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)



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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Aerosol insecticide formulation composition. The aerosol insecticide formulation shall conform to the composition of table I (see 4.2.3.1). No other ingredients shall be used in the aerosol insecticide formulation. The aerosol insecticide formulation shall pass the d-phenothrin content test when tested as specified in 4.2.4.1.

TABLE I. Composition of aerosol insecticide formulation

| Ingredient | Percent by weight |
|---|---|
| d-Phenothrin [3-phenoxybenzyl d-cis and trans 2,2-dimethyl-3-(2-methylpropenyl) cyclopropanecarboxylate], minimum* | 1.92 |
| Dichlorodifluoromethane conforming to type 12 of BB-F-1421 | 49.0 ± 1.0 |
| Trichloromonofluoromethane conforming to type 11 of BB-F-1421 | 49.0 ± 1.0 |

*The d-phenothrin shall contain a maximum of 25 percent by weight cis isomer and a minimum of 75 percent by weight trans isomer. The contractor shall certify that the isomer content requirement has been met.

3.2 Net weight of contents. The filled aerosol can shall contain 12.0 (+0.2 or -0) ounces of aerosol insecticide formulation when tested as specified in 4.2.4.2.

3.3 Aerosol can and valve-actuator. The aerosol can shall be a metal can conforming to type IX, class 2 or 3 of PPP-C-96. Coating of the can exterior shall be in accordance with plan B, with side seam stripe of PPP-C-96 and be forest green color in accordance with color 34079 of Fed. Std. No. 595. The aerosol can shall have a nominal capacity of 12 fluid ounces. The aerosol can shall be supplied with a valve-actuator system having the characteristics specified in table II. Each aerosol can shall also be supplied with an additional actuator and a snug-fitting, 5-inch long plastic extension tube attached to the side of the can. The extension tube shall have a nominal 0.040-inch inside diameter. The valve system shall be of the vertical action "sliding" gasket, valve type and shall conform to one of the following three models:



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(a) Precision Valve Corporation
Valve: N,N No. 1
Standard actuator: Kosmos buttons, 0.018 inch, Aqua (non MB)
Additional actuator: 11-5038
Mounting cup: 12-1230 (flowed-in gasket)
Stem: 04-1210-04, 0.013 inch diameter orifice
Gasket: 05-0310, "Buna N"
Spring: 06-6010, stainless steel
Valve body: 07-1901, 0.080 inch inside diameter
Dip tube: 09-2010, notched PVC, proper length to empty container contents

(b) Seaquist, Division of Pittway Corporation
Valve: Seaspray NS-31
Standard actuator: Excel 200 Power Mate (Polypro) No. 1205-0548-13 (0.013 inch) or No. 1205-0548-16 (0.016 inch)
Additional actuator: No. 0100-12180-40
Mounting cup: regular 1 inch (flowed-in gasket)
Stem: 0.016 inch diameter orifice
Gasket: neoprene
Spring: 0.023 inch diameter wire, stainless steel
Valve body: 0.080 inch inside diameter
Dip tube: proper length to empty container contents

(c) Newman-Green Ltd.
Valve: R10-118-4-9/16 (see dip tube)
Standard actuator: 130-27-18
Additional actuator: 102-27-40
Mounting cup: regular 1 inch (flowed-in gasket)
Stem: 0.016 inch diameter orifice
Gasket: neoprene
Spring: stainless steel
Valve body: 0.080 inch inside diameter
Dip tube: proper length to empty can contents

TABLE II. Actuator-valve characteristics

| Characteristic | Requirement | Test paragraph |
|---|---|---|
| Delivery rate at 80°F | 1.05 ± 0.30 g per second | 4.2.4.3 |
| Particle size distribution: | | 4.2.4.4 |
| Up to 30 micrometers, minimum | 80.0 percent | |
| 30 to 50 micrometers, maximum | 20.0 percent | |
| Over 50 micrometers | 0.0 percent | |
| Leakage | No evidence of leakage | 4.2.4.5 |



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3.4 Protective cap. The actuator-valve shall be protected against actuation by a protective cap as specified in PPP-C-96. The cap shall provide a clearance of no less than 1/8 inch directly over the actuator-valve. The cap shall provide a snug fit and shall completely cover the end of the aerosol can containing the actuator-valve assembly.

3.5 Workmanship. The aerosol insecticide formulation shall be mixed by a suitable process to produce a homogeneous mixture. The aerosol can shall be clean and free of any defects which would make it unsuitable for its intended use.

3.6 Labeling. Labeling data shall be legibly lithographed on the aerosol can with weather-resistant noncorrosive ink. Paper labels shall not be used. Each container shall be labeled in compliance with 40 CFR 162 and 49 CFR 171 to 179. Labels shall be approved and registered in accordance with 40 CFR 162. As a minimum, each container shall be labeled with the following information:

(a) The name and percent by weight of each active ingredient
(b) The name and percent by weight of each inactive ingredient
(c) National stock number, specification number, lot number, and date of manufacture
(d) Manufacturer's name and address
(e) EPA registration and establishment number
(f) Net weight of content in ounces
(g) d-Phenothrin cis/trans isomer ratio
(h) Storage and disposal data
(i) Precautionary information indicating hazards to humans and domestic animals, environmental hazards, and physical and chemical hazards
(j) Directions for use as follows:

FOR USE BY DOD PERSONNEL ONLY

FOR PUBLIC HEALTH, PLANT PROTECTION, AND QUARANTINE PESTS

GENERAL CLASSIFICATION

Directions for use

It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

DIRECTIONS FOR USE ON AIRCRAFT

To kill flies, mosquitoes, gnats, and caribbean fruit flies: Use to disinsect aircraft while in flight or while on ground prior to takeoff. Use at least 30 minutes prior to the first landing at a US port. Spray 10 grams (10 seconds) per 1,000 cubic feet of space. Move arm from side to side in a sweeping motion while spraying. Spray all spaces within the aircraft after loading is completed, crew and passengers are on board, and all doors, hatches, and ventilation openings are closed. Stop the ventilation system for at least 3 minutes after spraying.



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DIRECTIONS FOR USE IN BUILDINGS, VANS, SHIPS, AND IN TENTAGE

FOR INDOOR USE ONLY

FOR USE BY DEPARTMENT OF DEFENSE PERSONNEL ONLY

To kill flying insects (mosquitoes, houseflies, and gnats): Close space to be treated. Spray above head with a sweeping arm motion 3 feet from any surfaces. Spray 10 grams (10 seconds) per 1,000 cubic feet of space. Keep treated area closed for at least 30 minutes after spraying. Ventilate area before re-entry. Repeat as necessary.

To kill cockroaches: Remove standard spray actuator and attach additional actuator with an extension tube. Spray directly into cracks, crevices, and all places suspected of harboring cockroaches. Contact as many insects as possible with the spray. Special attention should be paid to hidden areas behind stoves, refrigerators, and food storage areas, and suspected hiding places.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the filled aerosol cans produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous.

4.2.2 Sampling.

4.2.2.1 For examination of filled aerosol cans. Sampling shall be conducted in accordance with MIL-STD-105, inspection level I, using an AQL of 2.5 percent defective.

4.2.2.2 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2, using an AQL of 4.0 percent defective.

4.2.2.3 For test. Sampling for test for net weight of contents, delivery rate at 80°F, particle size distribution, and leakage shall be conducted in accordance with MIL-STD-105, inspection level S-3 using an AQL of 2.5 percent defective. Three additional samples shall be selected for the d-phenothrin content test.



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4.2.3 Inspection procedure.

4.2.3.1 For composition of aerosol insecticide formulation. Weight records of the amount and lot number of each ingredient used to produce each lot of aerosol insecticide formulation shall be kept for the information of an inspector.

4.2.3.2 For examination of filled aerosol cans. The sample unit shall be one filled aerosol can. Sample aerosol cans shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Aerosol can corroded, bent or distorted
(b) Aerosol can not as specified
(c) Standard and additional actuator not as specified or missing
(d) Protective cap not as specified or not a snug fit
(e) Labeling incorrect, missing, or illegible
(f) Workmanship not as specified (see 3.5)

4.2.3.3 For examination of packaging. The sample unit shall be one filled shipping container of 24 aerosol cans, ready for shipment. Sample containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container or arrangement thereof not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged
(e) Fiberboard packing aids not as specified or missing
(f) Marking incorrect, missing, or illegible
(g) Unitization not as specified

4.2.3.4 For test. Sample aerosol cans taken in accordance with MIL-STD-105 in 4.2.2.3 shall be tested for net weight of contents, delivery rate at 80°F, particle size distribution, and leakage. The three additional samples taken shall be tested for d-phenothrin content. Failure of any of the sample aerosol cans to pass the d-phenothrin content test shall be cause for rejection of the lot represented.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 d-Phenothrin content.

(a) Apparatus. Use a gas chromatograph equipped with a flame ionization detector and a suitable integrating recorder or computing integrator. A 6-foot glass 80/100 "Chromosorb W" column with a general purpose stationary phase suitable for pesticides is recommended. Adjust the test conditions to obtain symmetrical peaks with a minimum of tailing and base line separation.



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(b) Preparation of standard solution. Weigh 0.065 gram of analytical standard (see 6.3) into a 10-milliliter volumetric flask, dissolve in ACS grade chloroform, and dilute to the volume mark with ACS grade chloroform. Add a small amount of sodium sulfate to insure dryness.

(c) Preparation of specimen solution. Cool the sample aerosol can to approximately 0°F. Punch several holes in the top to relieve pressure. Open the aerosol can top with a can opener and allow to warm to room temperature. When all propellants have evaporated, dissolve the residue in ACS grade chloroform, transfer to a 100-ml volumetric flask, and dilute to the volume mark with ACS grade chloroform. Transfer a 10-ml aliquot to a second 100-ml volumetric flask and dilute to the volume mark with ACS grade chloroform. Add a small amount of sodium sulfate to insure dryness.

(d) Procedure. Use a 10-microliter syringe to inject 2 or 3 microliters of the standard solution prepared in (b) into the gas chromatograph and record the peak area for each peak. Repeat this procedure two times. If the measured peak areas are within 5 percent of one another, repeat the procedure using the same size sample (2 or 3 microliters) of the specimen solution prepared in (c). Perform at least three gas chromatograph runs on the specimen solution. If peaks are not within 5 percent of one another, repeat the procedure until a standard reproducible measured peak area can be established.

(e) Criteria for passing test. Perform an analysis on each of the three sample aerosol cans and average the peak areas. The averaged peak areas shall equal or exceed the corresponding averaged peak areas of the standard solution.

4.2.4.2 Net weight of contents. Weigh the filled aerosol can to be tested in 4.2.4.1 to the nearest 0.01 gram. After emptying the can as specified in 4.2.4.1, dry and weigh the empty can to the nearest 0.01 gram. Calculate the net weight of contents by subtracting the weight of the empty can from the weight of the filled can.

4.2.4.3 Delivery rate at 80°F. Determine delivery rate in accordance with ASTM D3069 using a water bath maintained at 80.0 ± 0.5°F.

4.2.4.4 Particle size distribution. Determine particle size distribution in accordane with ASTM D3092.

4.2.4.5 Leakage. Spray the contents of a filled aerosol can for 3 seconds and pause for 2 seconds. Repeat this operation five times. Then totally submerge the aerosol can in a water bath maintained at 130° ± 2°F. Let the can remain in the water bath for 5 minutes. The emission of bubbles from the aerosol can during the 5 minutes shall be considered evidence of leakage.



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analysis. requirement through

5. PACKAGING

5.1 Packing. Packing shall be level B or industrial as specified (see 6.2).

5.1.1 Level B. Twenty-four filled aerosol cans shall be packed upright in a snug-fitting box conforming to type CF, class weather-resistant of PPP-B-636. The box shall be provided with full height partitions, top pads, and bottom pads fabricated from the same material as that used for the box. The box shall be closed as specified in the appendix to PPP-B-636.

5.1.2 Industrial. Aerosol cans shall be packed in accordance with MIL-STD-1188.

5.2 Unitization. Level B shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using a new, unused pallet conforming to type IV of NN-P-71. The load shall be secured to the pallet by strapping, shrink wrap, or stretch wrap. Industrial shipping containers shall be palletized to meet carrier acceptance and assure safe delivery of the packs at destination.

5.3 Marking. Level B shipping containers and unitized loads shall be marked in accordance with MIL-STD-129. Industrial shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Insecticide, d-phenothrin is intended for the uses listed in 3.6(j).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required (see 5.1)

6.3 Analytical standard d-phenothrin. Analytical standard d-phenothrin for use in 4.2.4.1 may be obtained from McLaughlin Gormley King Company, 8810 Tenth Avenue N, Minneapolis, MN 55427.

Custodian:

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Review activities:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-51484(EA) Insecticide, d-Phenothrin, Aerosol

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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